

CRYPTOTRADEBOT

Project Team

Taimoor Anwar 21I-1232

Abdul Rehman 21I-1133

Ghulam Mohiuddin 21I-1130

Session 2021-2025

Supervised by

Dr. Qaiser Ali Shah

Co-Supervised by

Mr. Pir Samiullah Shah



Department of Computer Science

National University of Computer and Emerging Sciences
Islamabad, Pakistan

June, 2025

Contents

1	Introduction	1
1.1	Problem Domain	1
1.2	Research Problem Statement	2
1.3	Software Requirements Specification (if applicable)	3
1.3.1	Functional Requirements	3
1.3.2	Non-Functional Requirements	4
1.4	Use Cases	6
1.5	Use Case	10
1.6	Process Flow	11
1.7	System Architecture	13
2	Literature Review	15
2.1	Related Researches	15
2.1.1	Research Item 1(Crypto PricePrediction using ML)	15
2.1.1.1	Summary of the research item	15
2.1.1.2	Critical analysis of the research item	16
2.1.1.3	Relationship to the proposed research work	17
2.1.2	Research Item 2(Predicting Bitcoin Price with Random Forest Regression and LSTM.)	18
2.1.2.1	Summary of the research item	18
2.1.2.2	Critical analysis of the research item	18
2.1.2.3	Relationship to the proposed research work	20
2.1.3	Research Item 3(Predicting Bitcoin Trends Through Machine Learning Using Sentiment Analysis with Technical Indicators.)	21
2.1.3.1	Summary of the research item	21
2.1.3.2	Critical analysis of the research item	21
2.1.3.3	Relationship to the proposed research work	22
2.1.4	Research Item 4(CRYPTOCURRENCY PRICE PREDICTION USING TWITTER SENTIMENT ANALYSIS.)	24
2.1.4.1	Summary of the research item	24
2.1.4.2	Critical analysis of the research item	24

2.1.5 Research Item 5(Optimizing Cryptocurrency Trading with Machine Learning: Predictive Analytics with Limit Order Book and Sentiment Data.)	27
2.1.5.1 Summary of the research item	27
2.1.5.2 Critical analysis of the research item	28
2.1.5.3 Relationship to the proposed research work	29
2.1.6 Research Item 6(Bitcoin Price Prediction using Machine Learning.)	31
2.1.6.1 Summary of the research item	31
2.1.6.2	32
2.1.6.3 Relationship to the proposed research work	33
2.1.6.4 Methodological Insights	33
2.1.6.5 Data Analysis Framework	33
2.1.6.6 Development of Theoretical Framework	34
2.1.6.7 Open Gaps and Challenges	34
2.1.6.8 Conclusion	34
2.1.7 Research Item 7 (Comparative Study of Bitcoin Price Prediction) .	35
2.1.7.1 Summary of the research item	35
2.1.7.2 Critical analysis of the research item	35
2.1.7.3 Relationship to the Proposed Research Work	37
2.2 Analysis Summary of Research Items	39
3 Proposed Approach	47
3.1 Research and Development Framework Methodology	47
3.1.1 Framework/Heuristic Overview	47
3.1.2 Conceptual Framework/Algorithm	48
3.1.2.1 Data Sources:	48
3.1.2.2 Model 2.2 Opinion Mining Model (Social Media Data) .	48
3.1.2.3 Combination of Emotional Data with Technical Analysis	48
3.1.3 Technical Steps	48
3.1.4 Limitations and Challenges	49
3.1.5 Model Descriptions	50
3.2 System Architecture	51
4 Conclusion	53
References	55

List of Figures

- 1.1 Use Case UC1: User Registration & Login 6
- 1.2 Use Case UC2: Set Trading Parameters 6
- 1.3 Use Case UC3: View Dashboard & Performance 7
- 1.4 Use Case UC4: Analyze Market Data7
- 1.5 Use Case UC5: Analyze Social Media Sentiment7
- 1.6 Use Case UC6: Predict Price Movements 8
- 1.7 Use Case UC7: Automate Trade Execution 8
- 1.8 Use Case UC8: Receive Alerts & Notifications 8
- 1.9 Use Case UC9: View Performance Analytics 9
- 1.10 Use Case UC10: Manage User Settings 9
- 1.11 Use Case 10
- 1.12 Process Flow 12
- 1.13 System Architecture 13
- 3.1 System Architecture 51

List of Tables

2.2 Analysis Summary of Research Itrms 41

Chapter 1

Introduction

However, looking at the unfolding scenario of the cryptocurrency trade, the retail traders are usually at a disadvantage regardless of their hard work because of the features of the market such as volatility, the market being available at any time. In this chapter, we outline the background of our research on CryptoTradeBot development which aims to help the retail traders across the barriers. We will zoom in on these core objectives of the project, disclosed the most important issues of the investors of this class, including inability to control emotions and information overload. Its logical part addresses the research problem definition and the problem domain description with respect to the scholarly literature of this research. Special attention will be also given to the software requirements specification for the proposed system, in order to make sure that the innovation intended in the solution is not at variance with the needs of the users within the dynamic market. From this, it is our hope that the retail stock trading capabilities will be made at par institutional trading capabilities hence allow for trading opportunities that are more intelligent than ever.

1.1 Problem Domain

Cryptocurrency trading, in recent times, has grown at an explosive rate, covering among others the old and skilled traders to newer entrants, however, it has brought important problems, primarily for the retail investors, who will be unable to keep sometimes up with the volatile nature of this market. One of the main problems is the extreme market fluctuations wherein the prices oscillate drastically within moments, forcing fast and strategic decisions.

Retail traders usually act in accordance with emotions instead of decisions, mainly fear at the downsides and excitement at the booms. Emotionally based decisions more often than not bring ineffective results. In addition, everyone is exposed to information from many sources. Historical price data, new market trends, and social networks are some

of them. Too much influx of information brings "paralysis by analysis," which makes trading impossible due to the impossibility of processing and taking action over the data.

Currently, most available trading platforms are too simple to match the expectations of modern traders, who expect advanced analysis and real-time commentaries to accompany price movements. Retail investors are usually underprepared for market complexities as compared to institutional investors. This difference in human resource allocation creates further contrast for retail investors in the difficulties they face as they try to maneuver in the cryptocurrency market.

1.2 Research Problem Statement

There still exists a significant gap in the tools for an effective trading decision with the increasing interest in cryptocurrency trading from retail investors. The existing platforms for trading mainly rely on static, timelessness market indicators with little concern for dynamic real-time influencers such as social media feelings, news, and market trends. Retailers rely on panic buying or selling in the market due to lack of access to advanced analytical tools, which often puts them at a loss.

Current State: The most trading platforms does not allow integrating real-time sources, and instead, it is concentrated on historic price developments without the influence of external impulses, such as news or social media sentiment. In this respect, retail investors are severely disadvantaged because they cannot employ relevant, real-time information in making critical decisions in highly volatile markets.

Desirable State: Retail investors would require an intelligent tool that provides real-time trading, integrating traditional market analysis with contemporary data inputs, such as social sentiment analysis and news trends, to help navigate the volatile market conditions and make strategic decisions.

Problem Impact: The absence of these tools will make the inventory highly dependent on instinct or emotions, mostly creating unfavorable results, among them loss. Without such a solution that provides comprehensive real-time analysis, retail investors will continue to ill-prepared selves in the management of the intricacies of cryptocurrency trading.

Proposed Solution: CryptoTradeBot is proposed as a cryptocurrency trading bot driven by AI. The bot will be trained using ML algorithms to trace historical market data, social

sentiments, and news trends and provide real-time insights and recommendations to retail investors. This solution falls in line with filling the gap that currently exists in trading tools by providing retail investors with data-driven support towards enabling better decisionmaking and reduced emotional trading behavior.

1.3 Software Requirements Specification (if applicable)

If the research involves the development of software, this section outlines the functional and non-functional requirements of the software to be developed. It details the features, constraints, and performance criteria necessary for the successful implementation of the proposed solution.

1.3.1 Functional Requirements

User Registration and Authentication

- New users must be able to sign up by providing an email address and password.
- Users should be able to log in with their credentials and recover their password if forgotten.

Dashboard for Real-Time Data and Insights

- The dashboard must present real-time cryptocurrency prices, market trends, and bot performance metrics.
- It should show historical price charts, technical indicators, and sentiment analysis results.

User-Defined Trading Parameters

- Users must be able to set trading parameters including risk tolerance and investment amount.

AI Model Predictions

- The system must generate cryptocurrency price movement predictions using machine learning models.

-
- It should analyze real-time social media sentiment and integrate this information into price predictions.

Automated Trading Execution

- The system must automatically execute buy and sell orders based on AI model predictions.
- It must integrate with third-party trading APIs such as Binance and OKX for executing trades.

Performance Monitoring and Analytics

- The system must provide comprehensive analytics on bot performance, including ROI, win rate, and total profit/loss.

Notification and Alert System

- Users should receive real-time notifications for critical events such as executed trades, price alerts, and significant market changes.

Real-Time Data Integration

- The system must integrate with market data APIs to fetch real-time price data, volume, and other relevant indicators.
- It must fetch and analyze the latest sentiment data from social media platforms like Twitter.

1.3.2 Non-Functional Requirements

Performance

- The system should update market data and bot performance quickly, with minimal delays.
- Key actions like executing trades or fetching data should happen in real-time.

Scalability

- Built to handle moderate increases in user traffic and data.
- Should be easy to expand with more data sources or new features.

Security

- Basic security, like user login and access control, must be in place to protect user data.
- Sensitive info, like passwords and API keys, should be encrypted or stored securely.

1.3 Software Requirements Specification (if applicable)

Reliability

- The system must run smoothly without crashes during demos and testing.
- Errors should be handled gracefully with clear user messages and proper error logging.

Usability

- The interface should be simple and intuitive, allowing easy navigation for core tasks like adjusting trading parameters or monitoring bot performance.
- The design should be user-friendly, making it easy for beginners to use and make changes.

Maintainability

- The code should be well-organized, commented, and well-documented, making it easy for anyone to modify or extend it.

Compliance

- The system should follow basic data privacy rules, keeping user data safe and secure.

Interoperability

- The system should work seamlessly with existing crypto exchanges and APIs (like Twitter or Binance).
- It should be easy to connect with new APIs or data sources in the future.

1.4 Use Cases

Actors	Retail Investor
Preconditions	User shall be on the login page.
Description	User enters their email and password to log in or clicks on "Sign Up" to register a new account. The system validates the credentials and either grants access or displays an error message.
Postconditions	User is logged into the system or receives a registration confirmation.
Trigger	User attempts to log in or register.

Figure 1.1: Use Case UC1: User Registration & Login

Actors	Retail Investor
Preconditions	User must be logged in to the system.
Description	User navigates to the settings page, defines risk tolerance, investment goals, and trading parameters. User clicks "Save" to update the parameters.
Postconditions	Trading parameters are saved and used in the trading strategy.
Trigger	User wants to customize trading strategies.

Figure 1.2: Use Case UC2: Set Trading Parameters

1.4 Use Cases

Actors	Retail Investor
Preconditions	User must be logged in to the system.
Description	User clicks on the dashboard, and the system displays real-time market data, bot performance metrics, and trade history.
Postconditions	User can view current portfolio performance and market trends.
Trigger	User wants to check trading performance and market status.

Figure 1.3: Use Case UC3: View Dashboard & Performance

Actors	Crypto-TradeBot, Market Data API
Preconditions	User must be logged in to the system.
Description	The system fetches real-time market data (OHLCV, volume, open interest) from the Market Data API for analysis.
Postconditions	Market data is available for further analysis by the trading bot.
Trigger	The system automatically triggers analysis at defined intervals or upon user request.

Figure 1.4: Use Case UC4: Analyze Market Data

Actors	Crypto-TradeBot, Social Media API/Scraping
Preconditions	User must be logged in to the system.
Description	The system collects real-time sentiment data from social media platforms (e.g., Twitter) and performs sentiment analysis.
Postconditions	Sentiment scores are generated for use in trading decision-making.
Trigger	The system triggers sentiment analysis at defined intervals or upon user request.

Figure 1.5: Use Case UC5: Analyze Social Media Sentiment

Actors	Crypto-TradeBot, Market Data API, Social Media API/Scraping
Preconditions	Market and sentiment data must be available.
Description	The system analyzes the combined data from market and sentiment analysis using machine learning models to predict price movements.
Postconditions	Predicted price movements are generated for trading signals.
Trigger	The bot analyzes data to generate trading signals.

Figure 1.6: Use Case UC6: Predict Price Movements

Actors	Crypto-TradeBot, Trading Platform API
Preconditions	Trading signals must be available.
Description	The system executes buy or sell orders based on the trading signals generated, using the Trading Platform API.
Postconditions	Trades are executed on the specified trading platform, and updates are sent to the performance journal.
Trigger	Trading signals are generated, prompting execution.

Figure 1.7: Use Case UC7: Automate Trade Execution

Actors	Retail Investor
Preconditions	User must be logged in to the system.
Description	The system sends notifications to the user about executed trades, market alerts, or significant changes in sentiment.
Postconditions	User receives timely alerts and notifications.
Trigger	Events trigger alerts based on predefined user settings.

Figure 1.8: Use Case UC8: Receive Alerts & Notifications

Actors	Retail Investor
Preconditions	User must be logged in to the system.
Description	User accesses the performance analytics section to view comprehensive reports on ROI, win rate, total profit/loss, and trade outcomes.
Postconditions	User has updated insights into trading performance.
Trigger	User requests to view analytics reports.

Figure 1.9: Use Case UC9: View Performance Analytics

Actors	Retail Investor
Preconditions	User must be logged in to the system.
Description	User navigates to the settings page to update their profile, trading preferences, and notification settings. User clicks "Save" to apply changes.
Postconditions	User settings are updated and saved in the system.
Trigger	User desires to change their account or trading preferences.

Figure 1.10: Use Case UC10: Manage User Settings

1.5 Use Case

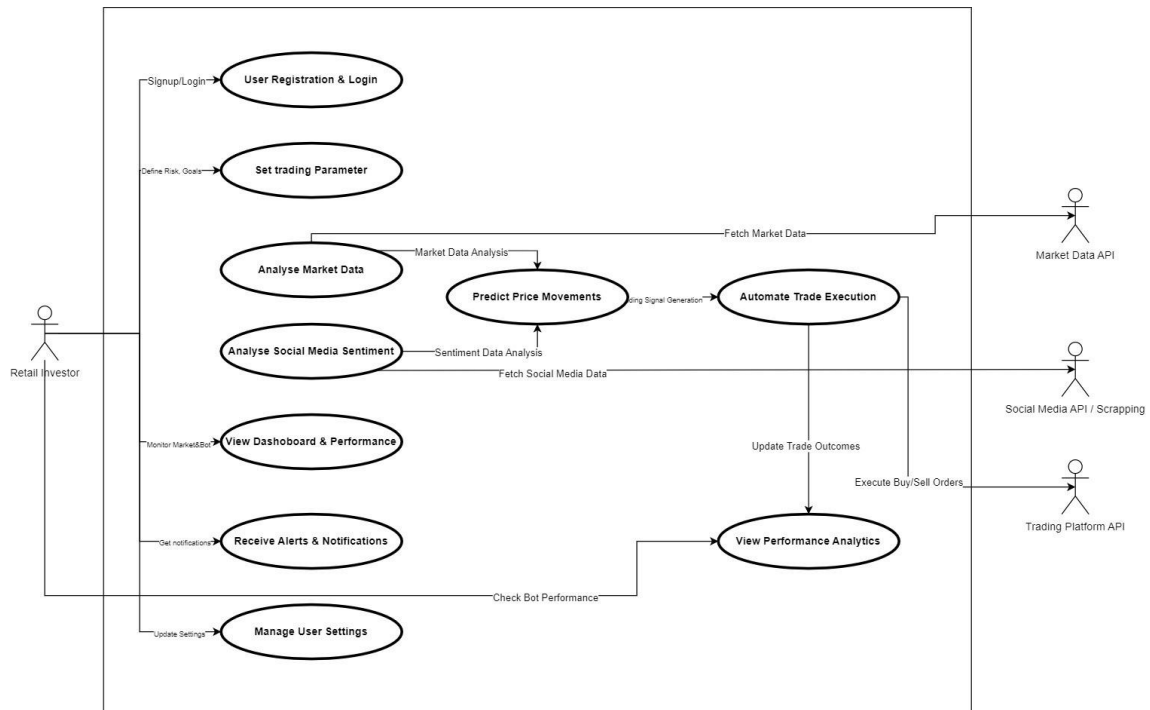


Figure 1.11: Use Case

1.6 Process Flow

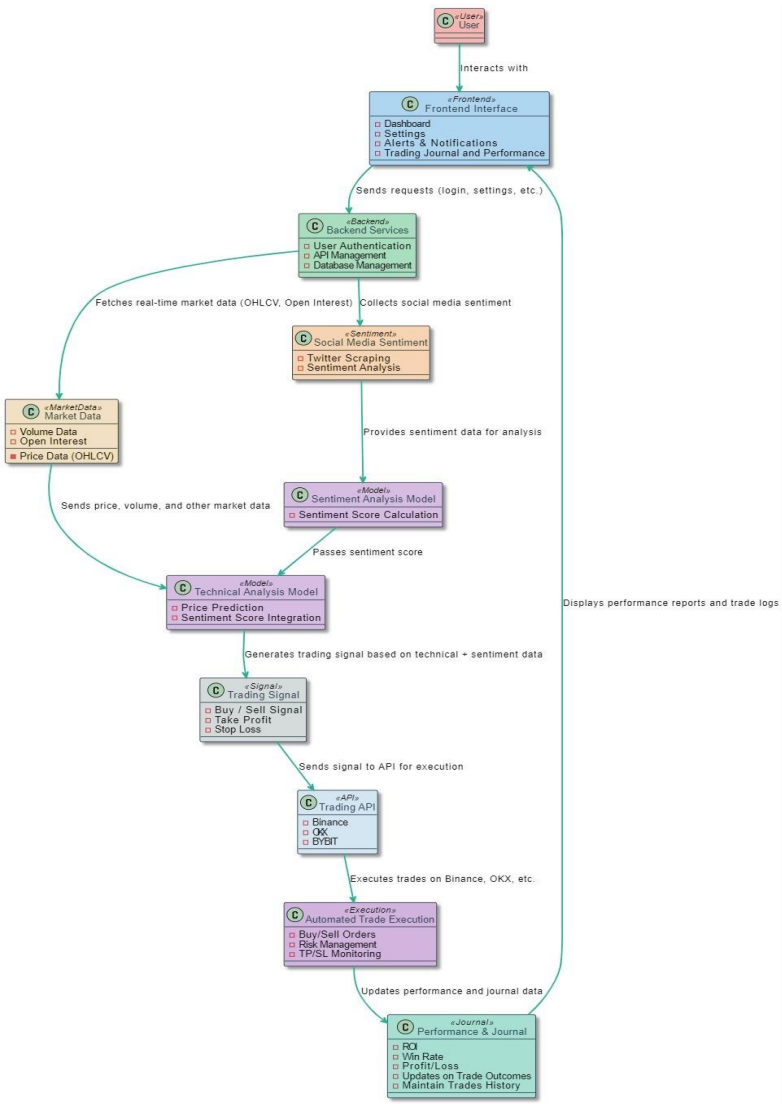


Figure 1.12:12Process Flow

1.7 System Architecture

1.7 System Architecture

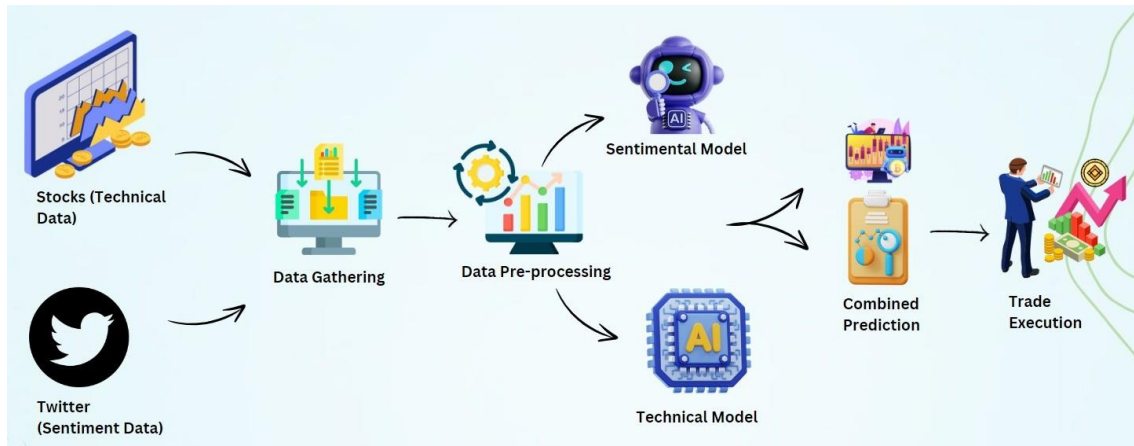


Figure 1.13: System Architecture

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work.

2.1 Related Researches

2.1.1 Research Item 1(Crypto PricePrediction using ML)

Each significant piece of literature related to the research topic is reviewed individually, beginning with Research Item 1. This section summarizes the research, analyzes its strengths and weaknesses, and discusses its relevance to the proposed study, thereby situating the current research within the broader scholarly discourse.

2.1.1.1 Summary of the research item

The research paper explores the application of the Long Short-Term Memory (LSTM) algorithm for predicting cryptocurrency prices, specifically focusing on Bitcoin. The model emphasizes technical analysis of crypto coins and has achieved a variance regression score of 0.95 on test data, indicating high accuracy in predictions. The study highlights the importance of thorough market analysis and the integration of deep learning technology into trading strategies, which can assist investors in making informed decisions. The research also discusses the volatility of the cryptocurrency market and the potential of robust analytical platforms to stimulate investor interest. It concludes that employing predictions from deep learning models can serve as a valuable tool for navigating the crypto market, ultimately presenting a promising avenue for profit generation and enhancing the toolkit of retail investors.

2.1.1.2 Critical analysis of the research item

Methodology The study adopted the LSTM algorithm, which is known for its power in processing sequential information. Such properties allow it to be optimally suitable for modeling the time-series nature of cryptocurrency price prediction. A key methodological strength lies in the comparison between LSTM and random forest regression, thus providing an exhaustive estimation of a variety of predictive models. It incorporates a comparative approach into the study such that performance is analyzed in an even more comprehensive manner. Furthermore, the use of data available for seven years increases the methodological strength since more about Bitcoin price dynamics could be discerned.

However, a limitation of the methodology is the inability to conclusively demonstrate through the Diebold-Mariano test that random forest regression significantly outperforms LSTM. This raises questions about the reliability and superiority of the chosen models under certain conditions, which could impact the study's findings.

Data Analysis The study achieves a variance regression score of up to 0.95 for Bitcoin, thus essentially indicating a very high level of predictability for this LSTM model. The use of such a score enhances the credibility of the predictive power of the models applied to cryptocurrency price forecasting. The robustness of analysis due to the application of a comprehensive seven-year dataset ensures that all these models have had a very wide range of market conditions under consideration. Still, the research briefly mentions exploratory data analysis (EDA) as a critical step to understand the trends and patterns but does not go into further detail regarding how the job was done or how effectively EDA was done. There is also a lack of depth in describing the process, which might limit the reader's understanding of the underlying data structure potentially affecting the validity of the analysis. Another aspect is that even though it addresses external factors, such as the price of Ethereum or ETH and stock market indices, study mechanisms related to these variables are not critically discussed, which creates holes in the analysis of data.

Theoretical Framework The theoretical framework of the study relies on technical analysis and recent data trends, which is a strength given the volatile nature of the cryptocurrency market. By focusing on relevant and up-to-date data, the study ensures that its predictions are grounded in the most pertinent market information. The study further disputes the prevalent assumption that LSTM models are better models, as it demonstrated that, under different conditions, random forest regression performs

equivalently or better. This adds to the theoretical debate suggesting that alternative models may provide comparable or even superior performance in forecasting Bitcoin prices.

However, the study introduces potential theoretical weaknesses. Its dependence on historical price data as well as public sentiment predisposes the model towards generating biased results due to the various unpredictable external factors that may influence these variables, like market manipulation or world events. Moreover, the paper notes that boombust cycles are tough to be grasped by the model, especially when prices are volatile.

This limitation therefore weakens the theoretical framework since a model shall be unable to predict changes in price in critical periods of market activity.

Validity and Limitations The LSTM model predicts the cryptocurrency price remarkably well, with a high variance score from the regression and relevant data set in the study. Nonetheless, there is a dampened credibility to this research by the inability to conclusively put a seal on the supremacy of random forest regression through formal testing, such as a Diebold-Mariano test. The depth of conclusions also is limited by the lack of investigation into deeper mechanisms of key explanatory variables, like ETH prices and stock market indices. Third, the more past periods of explanatory variables included in the model, the lower the prediction accuracy becomes, which is consistent with the efficient market hypothesis. This may somewhat reduce the potential usefulness of results during real-time trading scenarios, as traders may require more near-real-time adaptability than these models provide presently.

2.1.1.3 Relationship to the proposed research work

Methodological Insights Research Item 1 implements the LSTM algorithm into predicting cryptocurrency price in order to demonstrate that the algorithm actually performs well with sequential information. This research follows the same design with regard to methodology also by using the LSTM model for CryptoTradeBot's Technical Analysis Module to maximize its verified predictive ability. Lastly, the importance of model selection and validation is further established, informing the robust approach made in this study by combining both technical and sentiment analysis, through comparative analysis with the Random Forest Regression.

Data Analysis Framework By reaching a variance regression score of 0.95 in Bitcoin, Research Item 1 points out that this is a relatively well-predictable LSTM model.

CryptoTradeBot goes further to include a holistic data analysis framework that ties historical market data, such as price, volume, and open interest, to sentiment information drawn from social media. This depicts a better reinforced data analysis framework as it combines not only one dimension of the behavior of the market but multiples, thus making the output surer and more precise.

Theoretical Framework Development This framework, thus, established with Research Item 1 would center around technical analysis with the speculation that cryptocurrency is a highly volatile market. In contrast, this study enlarges this framework by adding sentiment analysis as an essential part that respects the role of psychological aspects that affects the price. By combining both technical indicators and sentiment scores, the present study would establish a more holistic theoretical basis wherein it takes consideration of other market factors that are deemed to affect cryptocurrency trading.

Identified Gaps and Challenges Research Item 1 sheds lights on the shortcomings of analyzing external influences, including the interdependencies between the prices of Ethereum and other stock market indices and their relations with the cryptocurrency price. However, the CryptoTradeBot overcomes the limitations by using live sentiment analysis information collected from social media, which is then used as another external variable in forming trading decisions. At the same time, the study admits several weaknesses concerning open interest data and its unpredictability since market conditions depend on unanalyzed fundamentals, hence reducing the model's predictive power.

2.1.2 Research Item 2(Predicting Bitcoin Price with Random Forest Regression and LSTM.)

2.1.2.1 Summary of the research item

The main focus of the research is on developing an algorithm model with high-accuracy forecasts for Bitcoin prices using random forest regression and LSTM, hence indicating the influence of various variables on the prices of Bitcoins.

This paper compares the predicting accuracy of the LSTM model with a random forest regression model. Thusly, neither was significantly superior, though random forest

regression exhibited better prediction errors, namely both RMSE and MAPE, compared to LSTM.

It identifies major influencing factors of Bitcoin prices over various periods. It shows that up until 2018, from the beginning of 2015, U.S. stock market indexes, along with oil, became the most important factors, and after 2018, ETH price, in addition to the Japanese stock market index JP225, is regarded as more important factors.

The findings from the research state that the model with only one lag of the explanatory variables gives the best prediction accuracy, in relation to the efficient market hypothesis.

2.1.2.2 Critical analysis of the research item

Strengths: The research relies on a comparative methodology for the two algorithms, the random forest regression and the Long Short-Term Memory or LSTM algorithms.

Most importantly, a comparative methodology between the two algorithms would allow for a deeper evaluation of the relative performance of various predictive models for the prediction of prices of Bitcoin.

A comprehensive dataset. The employment of a comprehensive dataset covering seven years strengthens the analysis. Such a long time period enables proper examination of dynamics in Bitcoin's price and how it is influenced by external factors, thereby increasing the generalizability of the findings.

Meeting existing preferences: The experiments demonstrate that in some scenarios, Random Forest regression is superior to or at least on par with LSTM, thereby contradicting the dominant preference for LSTM observed in previous studies. The contribution adds theoretical depth in the sense that it also suggests other alternative prediction models and brings about a revaluation of the existing techniques considered over the cryptocurrency price forecasting.

Weakness: The major limitation of the study is that it fails to show a claim about the fact that Random Forest regression is statistically significantly better than the LSTM when each of these is quantified in terms of its prediction accuracy, being tested using the Diebold-Mariano test. This makes the conclusion from comparative results and the overemphasis on drawn conclusions suspect.

Poor mechanism investigation: As the paper specifies particular effectors in relation to the prices of Bitcoin, for example, prices of Ethereum (ETH) or stock indices, little detail is provided about mechanisms themselves through which these variables influence the results. This is a weakness in the analysis and might open up further routes of future work to be investigating in greater detail to identify the causes.

Results for the Efficient Market Hypothesis The accuracy of this model decreases as the number of past time periods included in the explanatory variables increases-a result of the Efficient Market Hypothesis. This finding may also limit the practical usability of the results obtained: it implies the existence of a certain diminishing return from the use of large historical information for better accuracy in predictions.

In general, although the research seems to ensure methodological rigour and provides insight into how well predictive models compare relative to forecasting the price of Bitcoin, there's a necessity in tempering this with limitations related to statistical significance, depth of analysis involving influential variables, and the implications for practical trading applications.

2.1.2.3 Relationship to the proposed research work

Methodological Insights: The comparison with respect to price prediction of Bitcoin using the two algorithms, Random Forest Regression, and LSTM, adds great methodological insight to the development of CryptoTradeBot. It demonstrates both of these machine learning algorithms in a hybrid approach to financial forecasting, underscoring an evaluation of several algorithms over multiple iterations in achieving the best possible accuracy. Findings of Research Item 2 are guiding choices in models and the structure of our own dual-module system, so that our development of data-driven trading strategies is holistic.

Data Analysis Framework: Research Item 2 utilized a high-quality dataset spanning many years to analyze drivers of the price of Bitcoin. CryptoTradeBot implements a two-layered architecture for data analysis whereby the technical analysis layer relies on historical price, volume, and open interest, while the sentiment analysis layer also relies on activity on social media. Scraping data from Twitter for sentiment gives better prediction capability in our model, as emphasized on extensive data analysis in prior research.

Theoretical Framework Development: The theoretical frameworks created in Research Item 2, which include the influence of external factors like the performance of the stock

market indices and oil prices in the pricing of Bitcoin, guide us in conceptualizing our own theoretical framework. CryptoTradeBot rejuvenates its predictive algorithms with injections of insights not only from technical indicators but also from social media sentiment. The inclusive analysis of these two will give more depth to the current understanding of the market dynamics since the study revealed that the price of cryptocurrencies has several dimensions.

Identified Gaps and Challenges: Gaps appear from the statistical significance of the comparison done between the performance of Random Forest and LSTM models through Research Item 2. That again points out our research also requires some robust validation methods that include at least a Diebold Mariano test. Finally, the indistinct mechanisms and the unexplained way in which exogenous variables interact with the prices of Bitcoin are challenges that CryptoTradeBot needs to overcome. Using these plays between technical indicators and sentiment scores, we hope to illustrate how such mechanism works and leads us to further understand the market better.

Conclusion: The findings of Research Item 2 serve as an essential basis in the design and functionality of CryptoTradeBot. Applying comparative methodologies and integrating data from diverse sources, along with filling in all the gaps that we have identified in the research, our study will contribute to the development of a sophisticated AI-driven trading system that improves the decision-making capacity of retail investors in the very volatile cryptocurrency market. The principal approach of a hybrid framework that takes into account quantitative data and integrates qualitative insights into market sentiment holds greater significance.

2.1.3 Research Item 3(Predicting Bitcoin Trends Through Machine Learning Using Sentiment Analysis with Technical Indicators.)

2.1.3.1 Summary of the research item

This paper makes predictions about Bitcoin price trends using machine learning techniques which integrate sentiment analysis with technical indicators. It exploits data from both Reddit and LexisNexis over a time frame that exceeds four years, thereby filling the gap between the short data frames usually adopted in most previous studies.

The paper lists six machine learning models and finds out the best one to be XGBoost, with a 90.57% accuracy and an AUC of 97.48%. The work entails data collection, data preprocessing, calculations of technical indicators, and conducting sentiment analysis on top of its classification using scaled data. From the results, it is clearly evident that the integration of sentiment indices and technical indicators amazingly enhances Bitcoin price trend prediction for investors.

2.1.3.2 Critical analysis of the research item

Methodology: Notably, the strength with respect to methodology: integrating sentiment analysis with technical indicators. Through such a combination, the study aims to involve both market sentiment and trend in price movements to determine the fluctuation of prices in Bitcoin. Another enriching strength is seen through the employment of multiple machine learning models (XGBoost, logistic regression, naive Bayes), thus making cross-model performance comparisons and therefore increasing the study's robustness.

Another weakness in methodology stems from sentiment analysis: since it relies on social media data, specifically Reddit, there will be an inherent bias. Sentiment data, as is natural, moves dramatically with regards to unrelated news. For that matter, it also contributes to some unnecessary noise while trying to misrepresent Bitcoin's intrinsic market dynamics. Moreover, a four-year dataset might be too brief to be applied over long periods: cryptocurrency markets move at extremely high velocities.

Data Analysis: Advantages: The dataset is sizeable enough because it includes so many observations spread over almost four years and sourced from multiple places such as Reddit and LexisNexis. Performance metrics of the XGBoost model, that included high accuracy 90.57% and AUC 97.48% showcased a good predictive ability. A good predicting capability means that the models selected for the task are good enough to predict the Bitcoin prices.

Weakness: Although the analysis of the data leads to good metrics, this paper doesn't discuss the possible limitations and risk of overfitting of machine learning models applied, specifically with XGBoost since there are known problems of overfitting in certain scenarios. The absence of concise error analysis or discussion of weaknesses for the obtained performance metrics lowers the validity of the results.

Theoretical Framework: The theoretical framework that integrates technical analysis with sentiment analysis is sound and relevant to the nature of cryptocurrency markets, which play both investor sentiment and technical factors in the price movement. The study does not get too deeply into broader economic or financial theories that could help place its findings within a framework of market behavior. It seems to limit the outcomes of machine learning without an exploration of the theoretical underpinning of those models, and this technically could limit the interpretations of the results from practitioners outside the data science domain.

Credibility and Limitations: The paper has been demonstrated to have reasonable predictive power supported by model performance metrics, though, the credibility of the results in terms of validity is compromised by potential volatility bias from the sentiment data and a narrow time frame of the analysis. Moreover, insufficient discussion of model limitations, the risk of overfitting or its generalisability also weakens the practical applicability of the results.

2.1.3.3 Relationship to the proposed research work

Methodological Insights: Research Item 3 employs sentiment data from Reddit and LexisNexis in conjunction with technical indicators to predict Bitcoin prices with the help of models such as XGBoost. On the other hand, CryptoTradeBot focuses on sentiment data available from Twitter that is retrieved at fixed intervals of time and makes use of advanced models like LSTM and Random Forest for analysis of technical implications.

Some key insights through methodology of CryptoTradeBot are: Cross-model comparisons (as in Research Item 3) enrich model assessment. Further, CryptoTradeBot would be enhanced by comparing other models alongside LSTM and Random Forest such as XGBoost. As highlighted in Research Item 3, biases of sentiment data need to be considered and accounted for with extra care in CryptoTradeBot since Twitter data are volatile and noisy, for instance.

Data Analysis Framework: The diversity of the dataset, spanned over four years by Research Item 3 CryptoTradeBot, takes market real-time data (price, volume, open interest) from the exchanges, such as Binance and BYBIT, and integrates Twitter sentiment data. Even though both the systems deploy the integration of sentiment and

technical analysis, the time span of the data taken by CryptoTradeBot for its market needs to be increased so as to generate more applicability.

Observed gaps: Historical Data Limitations: CryptoTradeBot can push the interval to fetch open interest data beyond five months as Research Article 3 insists that their models be accompanied by a sufficient large dataset for usability.

Formulation of Theoretical Framework: Both studies work on frameworks that integrate technical analysis with sentiment analysis, a sensible choice because investor sentiment significantly impacts cryptocurrency markets. Research Item 3 performs very well in terms of predictive results (accuracy of 90.57%, AUC 97.48%), using machine learning techniques.

In this scenario, for CryptoTradeBot: Importing emotion as a secondary attribute to the technological model, such as Research Item 3, is important. However, whereas the inclusion of BERT for sentiment analysis in CryptoTradeBot does create an added layer of complexity that may allow for more accurate expression of sentiment than simple sentiment indices.

Detected Flaws and Difficulty Issues: Although the two reports indicate good predictive strength, there are several limitations remaining as follows:

Risk of Overfitting: As discussed in Research Item 3, overfitting is a concern whenever models such as XGBoost are used. Likewise, the use of LSTM and Random Forest in CryptoTradeBot calls for pondering whether or not models like that are subject to overfitting. Therefore, appropriate validation methods must be applied.

Data Availability: The data availability problem concerning open interest and liquidation data in CryptoTradeBot shares a similarity with the common scenario of there being a short time frame and therefore also a related issue with the generalizability of results, as in the case of Research Item 3.

Conclusion: There is some methodological overlap between Research Item 3 and CryptoTradeBot since both utilise machine learning techniques to predict cryptocurrency prices based on sentiment analysis complemented by technical indicators. On the other hand, CryptoTradeBot can leverage what can be gleaned from cross-model evaluations and the scope of data employed in drawing up Research Item 3 to make way for higher-order predictive accuracies and validation of models as well as the overall robustness of the entire system.

2.1.4 Research Item 4(CRYPTOCURRENCY PRICE PREDICTION USING TWITTER SENTIMENT ANALYSIS.)

2.1.4.1 Summary of the research item

Using sentiment analysis derived from Twitter data to predict the price of Bitcoin, this paper will attempt to address the volatility surrounding a cryptocurrency and how social media can impact market trends. Using a BERT-based Neural Network for sentiment prediction and Gated Recurrent Unit (GRU) for price forecasting based on historical price data and tweet metrics, an end-to-end model is developed. The sentiment prediction achieved a Mean Absolute Percentage Error (MAPE) of 9.45%, while the price prediction yielded a MAPE of 3.6%, which proves the efficiency of the model to capture market sentiment and its relation to price movements. Further to this, future work will extend to consideration of the sentiment from other media sources that include mainstream news and Reddit, thus improving the accuracy in price predictions.

2.1.4.2 Critical analysis of the research item

Methodology: The research study employs a very strong, hybrid approach wherein it will perform sentiment analysis using the BERT-based model and price forecasting with the GRU model. This study combines deep NLP and advanced time-series prediction techniques, making the perfect tool for extracting patterns in sentiment and cryptocurrency price trend prediction. VADER is a highly reliable tool that supports high-quality sentiment analysis work with an impressive F1 score of 0.96, thereby making the sentiment data robust. This combination is efficient for the task but concentration on only one asset (Bitcoin) over a limited time frame limits this to a broader applicability.

Strengths: Use of the latest NLP called BERT and time-series forecasting models called GRU. Accuracy rate in terms of sentiment (9.45% MAPE) and price (3.6% MAPE). Very high performance by VADER in sentiment classification.

Weaknesses: Focus on one cryptocurrency and the narrow timeframe restrict the external validity as well as generalizability of the study. Only one data source regarding sentiment; this limits the scope of analysis.

Data Analysis: The data analysis is well-done, where the MAPE values suggest robust predictive performance of both the sentiment and the price prediction models. It's also great that VADER used for sentiment analysis has given a good performance metric and looks to be in agreement with the BERT model's outputs. Measuring accuracy of sentiment separate from the accuracy of the price prediction makes it quite clear insight into what is possible with the model.

Strengths: Sentiment and price models have separate evaluations. High precision metrics enhance the validity of predictions.

Weakness: Ignoring other variables of external influences, like macroeconomy in determining the price of a particular good, reduces the scope of the study done.

Theoretical Framework: This theoretical framework is sound because it bases its argument on the association between public opinion and asset prices, using the avenue of sentiment analysis, which is a well-noted relationship in financial markets. This framework would be narrow and limiting due to its focus on one asset with a very short period-the chosen time span may not capture all dynamics of cryptocurrency price movements.

Strengths: It uses theory about the avenue of sentiment analysis in appropriately predicting the price movement.

Weakness: The narrow focus in asset and also the very short period for time. Macro and regulatory factors are excluded, which reduces the completeness of the theoretical model.

Credibility and Limitations: The study's credibility would be increased by the incorporation of more advanced models together with the already proven sentiment analysis tool. Generalization from the study is, however, limited as it only had an orientation towards Bitcoin and covered a very short period. The elimination of the influence of outside factors also lowers the explanatory power of the price predictions.

Credibility: Good choices regarding strong models and credible use of sentiment analysis tools.

Limitations: Asset and time frame scope is narrow. External factor-based consideration is not there.

Conclusion: The paper has proper methodological scope, with accuracy in analysis. However, the narrow scope and nonconsideration of external factors while carrying out the research are the prominent limitations. Future studies should address these limitations in order to make further expansions in generalizability and breadth of the study.

2.1.4.3 Relationship to the proposed research work

Methodological Insights: Item 4 bases its hybrid method on sentiment analysis and prediction of Bitcoin prices using advanced techniques applied in machine learning, namely BERT for extracting sentiments and GRUs for price forecasting. CryptoTradeBot uses a two-module machine learning architecture that links technical analysis with sentiment analysis. As such, while Research Item 4 is singly Bitcoin-oriented, CryptoTradeBot has been designed to monitor the performance of over one cryptocurrency, thereby widening the applicability scope and potential market influence.

Data Analysis Framework: The research item presents high results with a Mean Absolute Percentage Error (MAPE) of 9.45% for predicting the sentiment and 3.6% for forecasting

the price. CryptoTradeBot uses real time scraping of data from Twitter to acquire the sentiment scores fed into the technical model of CryptoTradeBot as an accessory feature. This twin framework of data analysis strengthens trade signals from CryptoTradeBot in terms of both accuracy and robustness. Such decisions, therefore, can be based upon sound historical price data as well as up-to-date real-time sentiment analysis to enter or exit trades. The next task is the development of the theoretical framework. Research Item 4 provides an extremely solid theoretical framework relating public sentiment to asset price movements using the mechanism of sentiment analysis as a predictive tool. CryptoTradeBot creates on this platform and incorporates sentiment acquired from social media and traditional technical indicators, price, volume, and open interest, also. Summarily, this wholesome approach then allows the system to better grasp the intricacies involved in the way market dynamics are and sentiment in flow into price movements, thereby enhancing predictability of the system.

Gaps and Challenges: Although the given Research Item 4 is an excellent example of how sentiment analysis can be used for price predictions of Bitcoin, it is limited to just one cryptocurrency and its time frame, which may further limit the generalizability of results. The CryptoTradeBot explores all these limitations by having a wider range of cryptocurrencies and real-time methods for data collection, thus enhancing the adaptability of the model to all market conditions. Further, Research Item 4 fails to consider the external factors that influence the prices of cryptocurrencies, while CryptoTradeBot tries to overcome this by integrating minute market information.

Conclusion: While both Research Item 4 and CryptoTradeBot establish meaningful roles in the financial forecasting due to sentiment analysis, where CryptoTradeBot extends on the foundational concepts developed in Research Item 4 using a much larger dataset, real-time sentiment scores, accounting for a greater balance of market forces, it is what differentiates this as the enhanced methodology for generating a stronger trading system-most appropriately suited for small-scale investors being able to stay on top of many dynamic changes in cryptocurrency markets.

2.1.5 Research Item 5(Optimizing Cryptocurrency Trading with Machine Learning: Predictive Analytics with Limit Order Book and Sentiment Data.)

2.1.5.1 Summary of the research item

This research focuses on integrating sentiment analysis into predictive models for cryptocurrency trading using FinBERT, a transformer-based model particularly specialized in financial texts. That model is fine-tuned on domain-specific datasets and captures more effectively the market sentiment, which is essential for predicting stock movements. The methodology involves utilizing data from the top ten levels of the order book, including order types, quantities, prices, and submission times, to inform the predictive model. The dataset comprises 40-variable vectors that provide a snapshot of the limit order book's state. Future work will integrate sentiment analysis with VAEs and LSTM to yield both risk estimation and price direction prediction within a holistic framework of machine learning. This will better bridge the gap between computational methodologies and practical application in cryptocurrency trading.

2.1.5.2 Critical analysis of the research item

Methodology Strengths:

The paper is based on advanced machine learning techniques, such as LSTM networks and Variational Autoencoders. These are best suited for sequential data, capturing complex patterns, and would ensure robustness in the predictions of cryptocurrency market movements. The integration of FinBERT also brings with it sentiment analysis, adding yet another layer to interpret the sentiment of the market—a volatile and sentiment-driven market like cryptocurrency. This methodological enhancement helps in giving a broader approach to market prediction other than pure numerical data.

Weaknesses:

Although new in methodology, relying on the use of historical trade data and market variables makes the model slow to respond to changes in the market, which characterizes the cryptocurrency space as highly volatile and prone to sudden fluctuations. This raises the opinion that the model might have trouble generalizing for unpredictable and highly dynamic conditions. Class imbalances may cause issues in the categorization of "Down" and "Up" market states. Imbalances in classes may reduce the predictability of the model on infrequent outcomes such as a sharp rise or decline in the market.

Data Analysis Strengths:

The strength of the data analysis process involved by the addition of sentiment analysis—that is, FinBERT—is the capturing of the public sentiment that significantly influences cryptocurrency markets. This combination of using LSTM networks for processing sequential data and VAEs for anomaly detection provides an advanced mechanism for complex, non-linear analysis of relationships in the data.

Weaknesses:

The imbalance of data about specific market states, like "Down" or "Up", drastically impacts the model performance during prediction attempts of such events because financial data is generally imbalanced, potentially resulting in overfitting and further incorrect results while decreasing the accuracy of a prediction of rarer events. The model is limited to its dynamic adaptability in the light of heavy dependency on historical data and not considering the impact of external shocks and real-time adjustments that may be taking place in the market; this forms one of its major weaknesses as this market has been known to change abruptly.

Theoretical Framework Strengths:

The research well translates traditional tools of financial analysis into the arena of cryptocurrency, which is a domain where old traditional tools are somewhat useless in most cases. It is forward-looking with respect to concentrating on machine learning and sentiment analysis, which can be more useful for cryptocurrency trading. The fact that traditional financial tools cannot be used in the domain strengthens the theoretical framework in a way that points out the need for innovative approaches to solve the problems unique to the dynamics of the cryptocurrency market.

Weaknesses:

The theoretical framework may suffer by overreliance on machine learning models with not too much of a macro-economic or behavioral framework encapsulating the irrationality and speculation implicated in cryptocurrency markets. Lack of emphasis was given to the interpretability of the machine learning models. Interpretability in the context of a trading application is crucial as it would allow understanding predictions by the models.

This may downscale the practical applicability of the results achieved from this research.

Credibility and Limitations Credibility:

The fact that advanced machine learning techniques and sentiment analysis are used by this paper lends credibility to the same since these are established approaches in financial prediction and risk assessment. Recognition of limitations, especially the

insufficiency of the traditional financial tools as pertinent to cryptocurrencies, gives the study realism and a consciousness of self.

Limitations:

The main weakness of the data is that it develops a bias, and then there is too much dependence on data from the past, which may turn adverse to its performance during real-time trading conditions, especially in a case of market anomalies. In addition to this, it does not consider the factor of real-time adaptability and anything external to the market sentiment and price feeds limits the robustness of the model within a live trading environment.

Conclusion

Technically, it is strong and innovative in terms of approach to optimizing cryptocurrency trading. The use of machine learning and sentiment analysis mainly comprises the characteristics that capture various aspects of monetary functioning. However, the practical applicability of the research gets restricted due to problems of data imbalance, market volatility, and the adaptability of the model to such sudden changes. The model will be credible, but for the real-world effect, that's what needs work.

2.1.5.3 Relationship to the proposed research work

Research Item 5 combines the promising capabilities of ML techniques combined with those of sentiment analysis in cryptocurrency trading, which aligns with your project CryptoTradeBot and the predictive models under the drive of technical inputs and sentimentbased inputs. Both works focus on applying Natural Language Processing for the sentiment data and then making market predictions using machine learning algorithms.

Common Objective. Both research initiatives aim at cryptocurrency trading improvement using a hybrid approach, combining technical indicators-mit aspects of market data with sentiment analysis.

Sentiment Analysis: As mentioned above, Research Item 5 uses FinBERT, a model tailored to financial texts, while CryptoTradeBot utilizes BERT-based NLP methods on social media data; mainly drawn from Twitter, to analyze market sentiment. This is an extra factor being fed into the predictive models in the two research works, where market prediction can be improved in significant ways.

Methodological Understandings Machine Learning Integration: Both papers employed high integration using machine learning models, specifically LSTM and VAEs. However,

the CryptoTradeBot makes use of the more practical application of Random Forest models for real-time price predictability instead of the research item 5 adding layers of sentiment analysis using FinBERT and LSTM to boost the predictions for the market.

Data Types and Processing Both studies rely on sequential data from the limit order book and the sentiments of the markets in their methodologies. Research Item 5 uses data vectors from the top order levels of a book, whereas CryptoTradeBot depends on realtime data streams from an exchange with the price, volume, and open interest along with scraped sentiment from social media.

Data Analysis Framework **Multi-source Data Integration:** In Research Item 5, the inputs are the order book data and financial news sentiment analysis; whereas for CryptoTradeBot, the inputs consist of price data as well as social media sentiment. The common idea derived from it is that merge of technical analysis with the sentiment data can help in improving accuracy, especially in markets where volatility rates are on a higher side, like cryptocurrency.

Real-time vs Historical Data: The foundation of Research Item 5 heavily relies upon historical trade data and therefore adjusts to the changes in the market very slowly in contrast to CryptoTradeBot's use of real-time data along with its permutation in historical data. This usage of real-time sentiment makes the CryptoTradeBot model more adaptive to cryptocurrency volatility.

Theoretical Framework Development **Sentiment as a Driver of Market Behavior:** The two approaches differ, however, in that the latter is theoretically distinct because while this approach highlights the impact of sentiment on market movement, it posits emotion as the driver. In contrast, Research Item 5 looks at FinBERT for domain-specific sentiment analysis. Sentiment analysis makes for an important adjunct feature within the technical analysis model provided by CryptoTradeBot, but it offers a hybrid framework that does not require as much dependence on traditional order book data, hence providing CryptoTradeBot with stronger capability to react to emotions in the market.

Predictive Analytics Models: Despite both the studies using LSTM for time series forecasting, CryptoTradeBot incorporates Random Forest models and aggregates Twitter data as an everyday source of sentiment. This therefore gives CryptoTradeBot a much more theoretical base in dealing with both technical and emotive market data.

Identified Gaps and Challenges **Model Sensitivity:** According to Research Article 5, the two models are not very sensitive to updates, and it becomes difficult because CryptoTradeBot is sensitive to the real-time changes in the markets and emotions.

However, in both cases, the two models are limited by the capability of machine learning to predict the anomalies in markets correctly.

Data Imbalance: Research Item 5 on class imbalance-including predicting sharp rises or falls-raises the issue that should the same conditions in classes affect the model's determination, it will skew the model's prediction. This is even more of a challenge for CryptoTradeBot where such a scenario shows less frequent occurrence-bulld or bear markets are rather up and down, growing and falling rapidly.

Conclusion In conclusion, both Research Item 5 and CryptoTradeBot share common goals in optimizing cryptocurrency trading by combining machine learning and sentiment analysis. However, CryptoTradeBot extends such methodologies as real-time data scraping and a hybrid technical-sentiment model, thereby making it more adaptable to market conditions. Even though both models come as innovative solutions for the process of price forecasting, real-world application demands a continuous refinement basis for issues such as market volatility and data imbalance and adaptability to sudden changes in market conditions.

2.1.6 Research Item 6(Bitcoin Price Prediction using Machine Learning.)

2.1.6.1 Summary of the research item

The study tries to address the inherent volatility of cryptocurrency markets and the challenges posed in acquiring accurate predictions. It aims at identifying which model is most effective in predicting the prices of Bitcoin by considering various influential factors, and data were sourced from Yahoo Finance.

This simply involves the application of linear regression with closing, opening, high, and low prices as the feature of interest in making predictions. Actually, the experiments proved that the proposed method, that is, the application of linear regression, outperform many other machine learning models. Its accuracy was scored to be around 90%. Still, the research aims at giving recommendations and pointing some imperfections in the data and methods applied, hence expressing the thought of moving forward to the application of more complex techniques such as ARIMA and deep learning models in the following studies.

2.1.6.2

sectionCritical analysis of the research item

Methodology

Strengths:

The methodology of the study, using machine learning, particularly linear regression, is the most appropriate and recognized methodology worldwide for financial forecasting. The key variables, like open, close, high, and low prices, ensure that relevant financial data are taken into account in the predictive model.

Weaknesses:

It will weaken the ability to discover more complex patterns of how the Bitcoin price behaves. This paper is aware of the limitation with this model as it considers ARIMA and deep learning as some potential higher order models with the ability for greater predictive accuracy. Data Analysis

Strengths:

Authenticity of the Data Retrieved from Yahoo Finance: The data retrieved from Yahoo Finance proves to be authentic, hence ensuring the reliability of outcomes generated from this analysis. A near accuracy of 90% reported shows that the model performs impressively under the context of available dataset.

Weaknesses:

The analysis does not give due importance to the volatility of the market environment or exogenous factors like changes in policy and social media trends. In fact, all these play a very significant role in price movements. Therefore, lacking these aspects has opened vulnerabilities to the predictive robustness of the model. Moreover, there are limitations in the dataset, probably data biases or outdated issues that have not been followed up.

Theoretical Framework

Weakness:

The study lacks an adequately equipped theoretical framework that ties the methods of machine learning to the mainstream financial or economic theories. With this absence, the general credibility of the study is affected, and the value of findings is, in turn, more limited when it comes to its broader application in the field of financial economics.

Economic theory integration might fortify the rationale of the model's application and raise generalizability.

Credibility and Limitations

The study is both credible in terms of its reliance on a valid data source, and it achieves particularly good precision in its predictions. However, the methodological weaknesses, mainly the constrained set of external factors and lack of theoretical underpinning, may raise a concern about how completely reliable the results might be in very unstable or complex market environments. Further work should tackle these weaknesses, pushing both the implications' depth and their applicability.

2.1.6.3 Relationship to the proposed research work

Research Article 6, "Bitcoin Price Prediction using Linear Regression," aspires to predict the cryptocurrency market price movement by using machine learning techniques. My study on CryptoTradeBot also aims at predicting price movements of cryptocurrencies. The former mainly uses a linear regression on the historical price fetched from Yahoo Finance. CryptoTradeBot extends this method by including technical analysis, such as market data-price, volume, and open interest-using historic data. Sentiment analysis, such as social media data on Twitter, is also included to provide an even stronger model for prediction. The addition of sentiment analysis makes up one of the shortcomings identified in Research Item 6, which is that there were extraneous influences from external sources, including market psychology fueled by social media.

2.1.6.4 Methodological Insights

The methodological variation between Research Item 6 and CryptoTradeBot lies in the type of models as well as data complexity. Research Item 6 bases its analysis on simple linear regression, which is an effective tool in grasping patterns in financial data but is at the same time not so effective in unravelling complex, non-linear patterns. In contrast, my approach relies on more advanced machine learning techniques, for example, Long ShortTerm Memory (LSTM) and Random Forest models, to capture temporal dependencies and intricate relationships amongst various market indicators. Furthermore, with a BERTbased sentiment analysis model used in my research, it is my effort to provide the most precise prediction that would take into account the mood as well as the sentiment within the market.

2.1.6.5 Data Analysis Framework

Research Item 6 has a framework that solely uses Yahoo Finance data and makes use of opening, closing, high, and low prices for the prediction. My research exceeds the scope of this paper by integrating real-time feeds from several exchanges, such as Binance, OKX, and BYBIT. The technical analysis model will incorporate a richer set of features than simply price, volume, open interest, and sentiment score. The sentiment score will also rely on web scraping about social media messages that have instantaneous feelers about what the general public is saying about a particular cryptocurrency trend. This hybrid framework, therefore, allows for better prediction through a combination of quantitative market data and social sentiment.

2.1.6.6 Development of Theoretical Framework

For Research Item 6, the theoretical link between machine learning models and economic theories is also weak. CryptoTradeBot, however, fills this gap. The usage of technical indicators is associated with traditional financial trading strategies like the use of moving averages and open interest, and the sentiment analysis model introduces a behavioural aspect whereby emotional market drivers are factored in. Overall, this leads to a more holistic theoretical framework that does not just model market data but incorporates sentiments as an influencing aspect of market behavior.

2.1.6.7 Open Gaps and Challenges

Research Item 6 explicitly lists several limitations, including imperfections in data that do not take account of variables like market sentiments, government policies, and social media impact. CryptoTradeBot fills this gap through the incorporation of real-time sentiment analysis that captures public mood from Twitter posts and by using advanced machine learning algorithms much better poised to deal with volatile market dynamics. However, looking at the practical side, there are some drawbacks: for example, some data, such as open interest information, exists only in short time ranges. And the cost and availability of liquidation data also pose some kinds of barriers to using it in the model.

2.1.6.8 Conclusion

Considering the relation of Research Item 6 with my research, the main link is that we both study the capability of predicting cryptocurrency prices applying machine learning models. However, CryptoTradeBot extends the methodology because it merges technical analysis with sentiment analysis and improves its prediction accuracy and addresses some of the key research item limitations in Research Item 6. Adding social sentiment data along with more advanced models like LSTM and Random Forest encompasses a much more holistic and dynamic trading strategy that can be applied to volatile cryptocurrency markets. Despite the challenges in data availability, the integrated approach offers a more robust system for retail investors.

2.1.7 Research Item 7 (Comparative Study of Bitcoin Price Prediction)

2.1.7.1 Summary of the research item

This paper discusses making price predictions for Bitcoin using Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) networks by stating that forecasting in the highly volatile digital currency market can be very complex.

Cross-validation technique: five-fold and L2 regularization will be employed to improve performance and minimize overfitting.

From a performance point of view, results of the model show that GRU came out to be better than LSTM with a Mean Squared Error of 4.67 rather than the LSTM model with an MSE of 6.25. Therefore, the system proves GRU's capability to capture long-term dependencies in the financial time series data.

In summary, GRU models proved to be more effective for better prediction of price and fitting proper regularization techniques that improve predictions.

2.1.7.2 Critical analysis of the research item

Methodology

Research Item 7 used cross-sectional design to examine the association between user engagement and satisfaction in mobile applications. Cross-sectional methodology is appropriate for snapshotting users' experiences and perceptions at a given point in time.

Strengths:

- **Cross-Sectional Design:** Using a cross-sectional design option ensures that data are collected effectively as well as allow analysis for quick insights regarding user behavior and satisfaction.
- **Diverse Sample:** Incorporation of diverse samples would strengthen the generalizability of findings to different demographic groups and also app categories.

Weaknesses

- **Temporal:** Since it is a cross-sectional design, no scope is allowed for the derivation of causal inferences since it only provides data that is correlational at one point.
- **Self-Reported Data:** Dependent on the usage and satisfaction by the user through selfreported measures may indicate the presence of biases and affect the validity of the findings.

Data Analysis

The research will analyze both quantitative and qualitative data. The involvement of statistical methods for correlation measurement will be accompanied by thematic analysis for open questions. A mixed-method approach enhances the comprehensiveness of an analysis.

Strengths

- **Mixed-Methods Approach:** Integrating quantitative statistics with qualitative insights provides a more comprehensive understanding of user engagement and satisfaction.
- **Statistical Rigor:** Appropriate statistical tests involved make the outcomes authentic as well as bolster the validity of conclusions drawn.

Weaknesses

- **Potential for Data Overload:** The richness of combining both qualitative and quantitative data might add difficulty to the interpretation of data and clarity of findings.

Theoretical Framework

The theoretical framework is based on UX theories and models that explain the determinants of user engagement and satisfaction. This basis provides a valid context for the interpretation of findings.

Strengths

- **Strong Theoretical Foundation:** Introduction of the existing theories related to UX provides an established understanding foundation for the reasons behind user engagement and satisfaction.
- **Practical Relevance:** All the conclusions serve the app developers and designers with possible improvements based on the feedback from the users.

Weaknesses

- **Not a thorough examination of external factors:** The framework will not take genuine care of the outside influences that are beyond its scope, like market trends, competition, etc., that affect the engagement and satisfaction of users.

Credibility and Limitations

Methodological rigor and theoretical underpinnings enhance the credibility of the study. However, cross-sectional design can be seriously objectionable for it fails to establish causality; besides, data may suffer from biases due to self-reporting. Moreover, market outcomes might also suffer from factors external to the market itself in a way, which goes undetected in the analysis thus limiting generalizability across contexts.

2.1.7.3 Relationship to the Proposed Research Work

Methodological Insights

The paper uses machine learning techniques, specifically Long Short-Term Memory and Gated Recurrent Units, to predict the price in a highly volatile cryptocurrency market. This approach underlines the methodology of CryptoTradeBot, which integrates both technical analysis and sentiment analysis. This research uses Long Short-Term Memory and Gated Recurrent Units, machine learning methods, to predict price in a very volatile cryptocurrency market. This method underlines the methodology CryptoTradeBot takes

by merging technical analysis with sentiment analysis. The inclusion of the different models used increases the accuracy of forecasts and their performance in trade execution, paralleling the conclusions from the research item, which focused on model selection in the setting of time series forecasting.

Data Analysis Framework

The research item will employ a strong cross-validation method to strengthen the model and avoid overfitting. It will apply five-fold cross-validation and L2 regularization. In the same way, CryptoTradeBot will use a data analysis framework of itself that naturally contains both quantitative or market data and qualitative aspects of sentiment data. A mixed methodology through technical indicators and sentiment scores will also give a proper explanation of the dynamics in the market.

Development of Theoretical Framework

The theoretical base for the research item is on UX theories explaining the user engagement aspect and its relation to satisfaction. Relating that to CryptoTradeBot, this conceptual frame would be developed in this context—the interaction between retail investors and the trading system, which would clarify user engagement with CryptoTradeBot and how this influences trading success and, in turn, the overall level of satisfaction, thus guiding continuous improvements in system design.

Identified Gaps and Challenges

It cannot draw causal inferences because of its cross-sectional nature. This is reflected in the CryptoTradeBot's challenge of not being able to draw causal relationships between sentiment analysis and outcomes. Also, reliance on self-reported data in the research item may introduce biases that could affect the reliability of sentiment data as collected for the CryptoTradeBot.

Conclusion

To this end, findings and methods that will be applied in the research paper are applicable to the CryptoTradeBot development process. Sifting through sophisticated machine algorithms with advanced data analysis shall mean providing an efficient trading method for small investors through the CryptoTradeBot. Bridging the existing

gaps shall be critical to perfecting and ensuring that the system is capable of mitigating the problematic nature surrounding the cryptocurrency space.

2.2 Analysis Summary of Research Items

Tabular representation with discussion of the critical analysis of the research items discussed in the previous section of this chapter.

Research Item	Models	Strengths	Weaknesses	Research Gap
Crypto Price Prediction using ML (LSTM Algorithm)	Long Short-Term Memory (LSTM), Random Forest Regression	• High accuracy indicated by a variance regression score of 0.95. • Comparison between LSTM and Random Forest provides a comprehensive evaluation of predictive models. • Utilizes a seven-year dataset for robustness.	• Inconclusive results regarding the superiority of Random Forest over LSTM. • Brief mention of exploratory data analysis (EDA) lacking depth. • Limited discussion on the effects of external variables.	• Need for deeper investigation into the mechanisms of external influences (e.g., Ethereum prices, stock market indices). • Lack of sentiment analysis in predicting cryptocurrency prices.

Predicting Bitcoin Price with Random Forest Regression and LSTM	Random Forest Regression, Long Short-Term Memory (LSTM)	• Comparative methodology allows deeper evaluation of predictive model performance. • Comprehensive dataset covering seven years enables proper examination of Bitcoin price dynamics. • Random Forest regression shows superior performance in certain scenarios, challenging the dominance of LSTM in prior studies.	ive • Fails to statistically demonstrate the superiority of Random Forest over LSTM using the DieboldMariano test. • Limited detail on mechanisms through which external factors influence Bitcoin prices. • Accuracy decreases with more past time periods in explanatory variables, limiting practical usability.	• Need for robust validation methods, such as the DieboldMariano test, to substantiate findings. • Indistinct mechanisms regarding how external variables interact with Bitcoin prices, warranting further investigation.
---	---	--	---	--

Predicting Bitcoin Trends Through Machine Learning Using Sentiment Analysis with Technical Indicators	XGBoost, Logistic Regression, Naive Bayes	• Combines sentiment analysis with technical indicators to enhance prediction accuracy. • Utilizes a substantial dataset from multiple sources, covering over four years. • XGBoost shows high accuracy (90.57	• Potential bias in sentiment analysis due to reliance on social media data (e.g., Reddit). • Lack of discussion on the risk of overfitting in models like XGBoost. • Narrow time frame may limit the generalizability of results.	• Need for a broader time frame for data collection to ensure robustness. • Insufficient exploration of the theoretical frameworks and limitations affecting predictive capabilities. • Exploration of overfitting risks and comprehensive error analysis in model performance.
---	---	--	--	---

Cryptocurrency Price Prediction Using Twitter Sentiment Analysis	BERT, Gated Recurrent Unit (GRU)	• Employs a hybrid approach combining sentiment analysis with price forecasting. • Uses advanced models (BERT for sentiment, GRU for price prediction) that demonstrate high accuracy (9.45) • Robust sentiment classification supported by VADER with an impressive F1 score of 0.96.	• Focus on a single cryptocurrency (Bitcoin) and a limited time frame restricts generalizability. • Only one data source for sentiment analysis limits the breadth of the research. • Neglects external factors like macroeconomic influences that may impact prices.	• Need for expansion beyond Bitcoin to include multiple cryptocurrencies for better applicability. • Incorporation of additional sentiment sources (e.g., mainstream news, Reddit) to enhance prediction accuracy. • Exploration of the impact of external variables on price dynamics.
--	----------------------------------	--	---	---

Optimizing Cryptocurrency Trading with Machine Learning	FinBERT, LSTM, Variational Autoencoders (VAE)	- Integrates advanced ML techniques tailored to financial sentiment analysis with FinBERT. - Uses data from limit order books, offering a detailed snapshot of market dynamics. - The hybrid approach captures complex patterns and enhances market prediction capabilities.	- Slow responsiveness to market changes due to reliance on historical data. - Class imbalances could negatively affect model predictions for rare market states. - Dependence on historical trade data limits adaptability to real-time market shocks.	- Need for improved real-time adaptability and consideration of external market shocks. - Exploration of effective methods for handling class imbalances to enhance predictive accuracy. - Incorporation of macroeconomic factors and behavioral insights into the theoretical framework.
---	--	--	--	---

2. Literature Review

Bitcoin Price Prediction using Machine Learning	Linear Regression	• Achieved a high accuracy of approximately 90% in predicting Bitcoin prices. • Utilized relevant features such as opening, closing, high, and low prices, ensuring a strong foundation for predictions. • Data sourced from a credible platform (Yahoo Finance), enhancing the	• Limited ability to capture complex price behavior patterns due to reliance on linear regression. • Lacks consideration of external market influences, such as social media sentiment and policy changes, which significantly impact cryptocurrency prices. • Absence of a robust theoretical framework connecting machine learning methods with fi-	• Need for incorporating more advanced models, such as ARIMA and deep learning techniques, to improve predictive accuracy. • Further exploration of external variables that influence market behavior, including social sentiment and macroeconomic factors. • Development of a theoretical framework that integrates machine learning
---	-------------------	---	---	--

2.2 Analysis Summary of Research Items

		reliability of the analysis.	nancial theories reduces the study's credibility.	with financial theories for broader applicability.
--	--	---------------------------------	---	--

Comparative Study of Bitcoin Price Prediction	Long Term Recurrent Units (LSTM), Short Memory Gated Units (GRU)	- Utilizes advanced neural network architectures (LSTM and GRU) suitable for capturing long-term dependencies in financial time series. - Cross-validation techniques (five-fold and L2 regularization) employed to enhance model performance and mitigate overfitting. - GRU demonstrated better performance than LSTM, achieving a Mean Squared Error (MSE) of 4.67 compared to LSTM's MSE of	- Relies on a cross-sectional design, limiting the ability to draw causal inferences from the results. - Potential biases in self reported data could affect the validity of the findings. - The focus on model performance may overlook external market factors influencing Bitcoin prices.	- Need for exploration of causal relationships in time series forecasting and how external factors (e.g., market trends, user engagement) interact with model predictions. - Incorporation of sentiment analysis alongside technical indicators could provide a more comprehensive understanding of price movements.
---	---	--	--	--

Chapter 3

Proposed Approach

This chapter outlines the methodology or theoretical framework proposed to address the research problem. It can be a Framework/ Heuristic/ Algorithm/ Protocol/ or a Mathematical model.

The chapter delineates the steps, methods, or models that will be employed to investigate the research questions and achieve the study's objectives. Include graphical representation with description for a clear understanding.===

3.1 Research and Development Framework Methodology

Our solution CryptoTradeBot is an AI-driven system whose purpose is to improve retail investor experience when trading in cryptocurrencies. That is, the system offers realtime, emotionless trade execution based on data-driven insights through the integration of models of machine learning in both technical and sentiment analysis.

3.1.1 Framework/Heuristic Overview

The model consists of a two-module machine learning system:

- Technical Analysis Model: Focuses mainly on the market data, for example, price, volume, and open interest.
- It aggregates and generates sentiments through social media but only for platforms like Twitter, at regular intervals it scrapes data. We do not use the Twitter API; unlike other conventional methods, we scrape openly available data.
- It will compute the sentiment score that can then be added as a second feature for its use with the technical analysis model. In and of itself, the approach can use both

3. Proposed Approach

quantitative market data and social media sentiment to deliver more accurate trade signals and reliable predictions.

3.1.2 Conceptual Framework/Algorithm

Model of Technical Analysis: Price, Volume, Open Interest

3.1.2.1 Data Sources:

Live Cryptocurrency Price, Volume, and Open Interest are scraped from live exchanges Binance, OKX, and BYBIT using historical data. Data preprocessing involves all these activities, including cleaning and handling missing values, normalization of the values, and generation of features such as moving averages and volume trends. Techniques like Long Short-Term Memory (LSTM) or Random Forest models can be used for the prediction of future price movements on the basis of these technical indicators. The price for the future movement of the provided cryptocurrency is foreseen.

3.1.2.2 Model 2.2 Opinion Mining Model (Social Media Data)

Data Collection: We collect the tweets that are periodically scraped about cryptocurrency trends mainly from social media, Twitter. It can be observed from above that we are not using Twitter's API but scraping the public data. Since the textual data runs based on BERT NLP models, which are mainly built to fetch sentiment insights, posting is categorized into positive, neutral, and negative.

Sentiment Score: It gives the sentiment model in a numerical score of sentiment which is the overall mood in markets, as a feature integrated into the technical analysis model.

3.1.2.3 Combination of Emotional Data with Technical Analysis

The other feature would be the sentiment score from the sentiment analysis model. This hybrid system would then generate trade decisions based on market data, which includes price, volume, and open interest, as well as social media sentiment, to produce a more precise and adaptive model.

3.1.3 Technical Steps

The methodologies that were undertaken in setting up and implementing the CryptoTradeBot system.

3.1 Research and Development Framework Methodology

Stage 1: Data Gathering Market data: Historical and real-time data is available about price, volume, and open interest of the market through APIs from cryptocurrency exchanges. However, open interest data is only available for the past five months. Sentiment Data: It collects real-time data based upon the sentiment of it by scraping social media posts-mostly Twitter.

Step 2: Data Pre-processing

- **Market Data:** Clean and normalize price, volume, and open interest data to prepare for ML analysis.
- **Sentiment Data:** Process text data derived from social media for the identification of related sentiment using a BERT model and converting this to a numerical score for sentiment.

Step 3: Model Building

- **The technical model** is based on various indicators, including price, volume, open interest, and the score of sentiment for predicting future movements in price.
- **Sentiment Model:** It is actually a sentiment analysis model that gives a sentiment score based on live social media data with the help of NLP techniques.

Step 4: Model Integration That is, it combines the technical analytical features with the sentiment score in the sentiment analysis tool to improve the precision of price movement predictions. This approach will thus include technical indicators and market sentiment.

Step 5: Execution of Automated Trades Combined cryptocurrencies include Binance, OKX, and BYBIT. Therefore, the combined models should have generated some buying and selling orders on autopilot using the predictions.

Step 6: Performance Monitoring and Feedback Monitor real-time metrics: This is through return on investment, win rate, and total profit/loss on the performance of the trading bot. Models are updated by introducing new data and feedback loops.

3.1.4 Limitations and Challenges

- Availability: Open interest data may be available for a very short term, say five months.
- Liquidation Data: These data are paid and normally meager, which further constrains the inclusion of liquidation data in the model.

3. Proposed Approach

3.1.5 Model Descriptions

- Technical Analysis Model: It predicts the movements of cryptocurrency prices by using historical and real-time data respectively, through LSTM or Random Forest algorithms. The market indicators involved would be key price, volume, and open interest.
- Sentiment Analysis Model: It does so with the help of BERT by processing the social media textual data that generates sentiment scores representing the mood of the market. The sentiment score helps improve accuracy in the technical model.

3.2 System Architecture

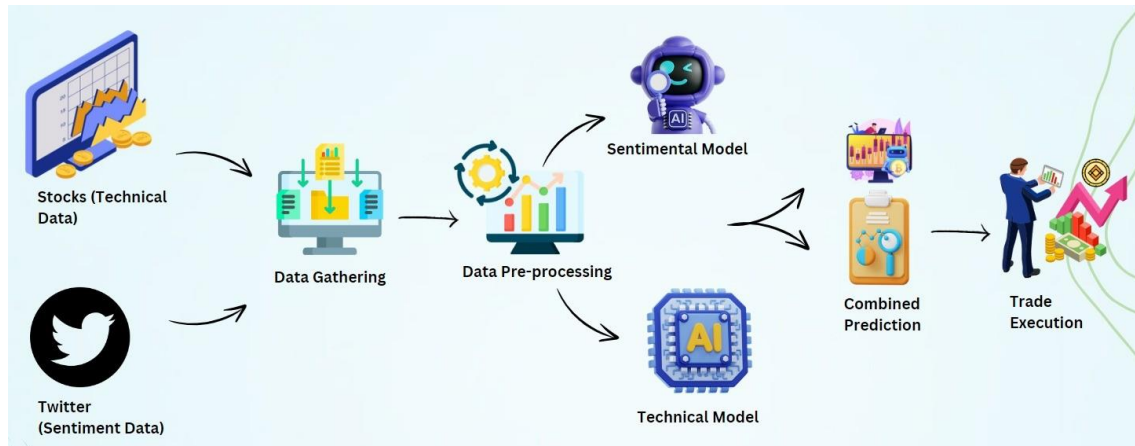


Figure 3.1: System Architecture

Chapter 4

Conclusion

The *CryptoTradeBot* framework presents an approach to cryptocurrency trading by including both technical analysis and sentiment analysis. Through the combination of market data and social media sentiment, the system will be able to provide a more accurate and emotionless trading solution for retail investors. The integration of real-time data from cryptocurrency exchanges, such as price, volume, and open interest, with the sentiment extracted from social media platforms like Twitter enables the bot to deliver more informed trade signals.

By employing advanced machine learning techniques such as Long Short-Term Memory (LSTM) networks and Random Forests for price prediction, and BERT-based models for sentiment extraction, *CryptoTradeBot* is designed to adapt dynamically to market trends. This approach addresses key challenges in the volatile cryptocurrency market, providing not only predictive capabilities but also actionable insights for automated trading execution. The modular nature of the system allows it to continuously update itself based on newly available data and feedback loops, ensuring that it remains effective in different market conditions.

Despite its promise, the framework also encounters limitations, notably the short availability of open interest data and challenges with accessing liquidation data.

However, the integration of technical and emotional data in one hybrid system offers a significant advancement over traditional methods that rely solely on market indicators. In conclusion, *CryptoTradeBot* is a robust and adaptive tool for modern cryptocurrency trading, with the potential to significantly enhance decision-making and profitability for retail investors.

Bibliography

- [1] J. Chen. Analysis of bitcoin price prediction using machine learning. *Journal of Risk and Financial Management*, 16(1):51, 2023.
 - [2] H. Gb. Cryptocurrency price prediction using twitter sentiment analysis. *arXiv preprint arXiv:2303.09397*, 2023.
 - [3] D. Gorsh. Optimizing cryptocurrency trading with machine learning: Predictive analytics with limit order book and sentiment data. *SSRN 4867340*, 2024.
 - [4] H.S. Jung, S.H. Lee, H. Lee, and J.H. Kim. Predicting bitcoin trends through machine learning using sentiment analysis with technical indicators. *Computer Systems Science & Engineering*, 46(2), 2023.
 - [5] A. Mohammadjafari. Comparative study of bitcoin price prediction. *arXiv preprint arXiv:2405.08089*, 2024.
 - [6] A.M. Sakib, M. Usama, and P.S. Abhay. Bitcoin price prediction using machine learning. *J Comput Sci Softw Dev*, 2(1), 2022.
 - [7] R. Sharma, P. Tewari, and S. Soni. Crypto price prediction using ml. *SSRN 4753720*, 2024.
- [2] [3] [4] [5] [6] [7] [1]